Enhanced Edge Information and Prototype Constrained Clustering for SAR Change Detection

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***Abstract*— The utilization of synthetic aperture radar (SAR) imagery for change detection can effectively circumvents the stringent limitations imposed by weather and lighting conditions, and is finding widespread applications in fields such as disaster monitoring and urban research. To address issues of edge blurring, severe noise interference and sample imbalance, an automated SAR change detection framework is proposed based on enhanced edge information and prototype constrained clustering. First, a gradient-based neighborhood ratio (GNR) is designed to reinforce the edge information of the difference map, facilitating robust differential information representations. Subsequently, to obtain accurate samples in an unsupervised manner, we have developed prototype constrained hierarchical clustering for preclassification. The quantity and quality of selected samples can be precisely guaranteed through the utilization of histogram analysis and prototype constraints. In the sample learning and prediction phases, a class-balanced noise-tolerant change detection network is proposed that combines focal loss and mean absolute error loss (MAEL), further tackling the sample imbalance issue, strengthening noise resistance and improving change detection accuracy. Comprehensive experimental results and analysis conducted on five benchmark datasets have validated the effectiveness and robustness of the proposed method.**

***Index Terms*— Change detection, class-balanced noise-tolerant network (CBNTNet), enhanced edge information, prototype constraints, synthetic aperture radar (SAR).**

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I. Introduction

CHANGE detection is a pivotal research and application focus in remote sensing. Its primary objective is to accurately extract change information through comparative analysis of images taken at different times of the same scene [1], [2]. Various imagery sources, including optical [3], [4], hyperspectral [5], synthetic aperture radar (SAR) [6], as well as combinations of different types of remote sensing imagery [7], have been extensively applied in practice. In comparison to other sensing modalities that have stringent requirements for lighting and weather conditions, SAR has garnered widespread attention due to its all-weather and all-day imaging capabilities, as well as its unique imaging characteristics. It effectively overcomes the limitations of Earth observation under harsh imaging conditions, and has been applied extensively in monitoring natural disasters and resources, such as flood mapping [8], [9], earthquake damage assessment [10], [11], forest fires detection [12], and sea ice monitoring [13], as well as urban studies [14], [15]. Particularly, with the continuous improvement in quality and quantity of remote sensing images, and the rapid development of advanced data processing techniques, there is an increasing demand for in-orbit intelligent and automated change detection methods. Accordingly, the integration of domain knowledge in SAR image change detection with deep learning models has become a mainstream research direction in this field.

Conventional SAR change detection usually involves three main steps: 1) image preprocessing; 2) difference map generation; and 3) difference map analysis [16]. Among these steps, the generation of the difference image serves as a fundamental component for SAR change detection, and its quality directly affects the final results. Consequently, numerous researchers have focused on developing various algorithms for difference map generation. The logarithmic ratio (LR) algorithm [17] stands out as a classic method, employing logarithmic operations to compress the difference space, thereby enhancing

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the accuracy of the subsequent difference map analysis. However, this algorithm is susceptible to noise interference. To address this, researchers have utilized neighborhood information to mitigate the influence of isolated noise. Inglada and Mercier [18] substituted the corresponding pixel value calculation with the neighborhood mean. Gong et al. [19] observed that solely using the neighborhood mean neglects the impact of heterogeneous pixels on the degree of difference, and proposed to incorporate the coefficient of variation to account for the degree of heterogeneity, resulting a neighborhood-based difference map. Zhuang et al. [20], [21] introduced a nonlocal theory to enhance the edge information of the difference map. Moreover, researchers have approached the problem from a fusion perspective, combining the respective strengths of various independent difference map generation algorithms. This includes algorithms that integrate neighborhood differences with single pixel differences [22], algorithms that combine Gaussian logarithmic ratio with LR [23], and methods that jointly use differential operators, LR, and logarithmic mean ratio (LMR) for difference map generation [24]. In summary, effectively suppressing the influence of speckle noise during difference map generation and improving the accurate representation of edge information are two critical courses of action in difference map generation.

The analysis of difference maps can be divided into two distinct approaches: conventional and deep learning-based methods. For its methodological simplicity and computational efficacy, conventional threshold selection paradigms and clustering analysis algorithms have been amalgamated. By leveraging the distribution characteristics of difference maps, researchers have developed various threshold segmentation methods, including the Gaussian expectation-maximization threshold [16] and the generalized Gaussian Kullback–Leibler threshold [25]. In contrast to threshold segmentation modalities, clustering algorithms offer an adaptive manner to transcend the confines of the specific histogram distribution prerequisites of the differential map. These include hard segmentation methods like the K-means algorithm [26], [27], and fuzzy-rule-based clustering algorithms. Typical examples include the fuzzy C-means (FCM) algorithm, the Markov random field-based FCM [28], reformulated fuzzy local-information C-means [29], multiobjective fuzzy clustering that breaks down the task into noise suppression and detail preservation subproblems [30], and contourlet fusion clustering [31].

With the advent and developmental refinement of deep models, numerous advanced difference map analysis algorithm have emerged. The primary strategy involves adopting the concept of hierarchical clustering [32], preclassifying difference maps in an unsupervised manner, utilizing pixels with easily identifiable categories to construct samples, and designing network models to learn from these samples and predict pixels with ambiguous change attributes [33]. These methodologies predominantly consider two aspects: 1) the quality of samples in the preclassification results and 2) the learning capability of the deep learning models. Factors affecting the quality of samples during the preclassification stage include the quantity, accuracy, and balance of the samples. Zhao et al. [34] adopted FCM for preclassification, and then randomly construct unchanged and changed samples for model learning. This approach guarantees sample balance. However, random selection may result in unstable training and predictions. Gao et al. [35] combined Gabor features with hierarchical FCM to randomly select a fixed proportion of changed pixels (CPs) and unchanged pixels (UPs) as samples, and it suffers from unpredictability due to its randomness and a predefined sample ratio issue. Furthermore, generative adversarial network (GAN)-based algorithms [36], [37] have been proposed to compensate for the deficiency in the number of unchanged samples, yet the accuracy of the samples generated by GAN remains to be ascertained. To address the noise issue, various classification models have been employed. Dong et al. [38] introduced the use of multiscale convolution and self-attention mechanisms, fully exploiting spatial neighborhood information for noise suppression. A dual-path denoising network was presented that integrates random label propagation to clean label noise [39]. Aiming to alleviate the effect of speckle noise, 3-D gray level co-occurrence matrix was also incorporated as spatio-temporal information into the convolutional neural networks [40].

Despite numerous effort devoted, SAR change detection still encounter the following three challenges. First, SAR images are inherently susceptible to noise, and noise present in any of the single-temporal SAR images will affect the difference map and the final change detection results. Second, accurate delineation of change region boundaries is difficult. Boundaries correspond to transition zones from changed to unchanged regions, where the degree of difference attribution is not distinct, making it difficult to be determined. Third, unsupervised ways of selecting samples suffer from class imbalance. In multitemporal SAR imagery, UPs often dominate the scene. Consequently, the number of unchanged samples constructed in an unsupervised manner usually far exceeds that of changed samples, thereby impacting model training and prediction. To tackle the aforementioned issues, this study proposes to combine the enhanced edge information and prototype constrained clustering for SAR change detection, termed as EIPCNet, as shown in Fig. 1. In the proposed EIPCNet, the gradient-based neighborhood ratio (GNR) is first computed to produce difference image and give prominence to changed areas. Then, a prototype constrained hierarchical fuzzy local-information C-means (PCH-FLICM) algorithm is developed to generate preclassified pseudolabels. We further construct a class-balanced noise-tolerant network (CBNTNet) trained with pseudolabeled bitemporal SAR image patch to derive final change detection results. Our main contributions can be summarized as followed.

1) The proposed EIPCNet leverages GNR to provide a better representation of difference information of edge boundaries, simultaneously compressing the overall difference space, while increasing the separability between changed and unchanged classes.

2) Taking into account the neighborhood information, we propose the PCH-FLICM algorithm for generating pseudolabeled samples from difference image. The incorporation of histogram and prototype constraints help enhance the stability

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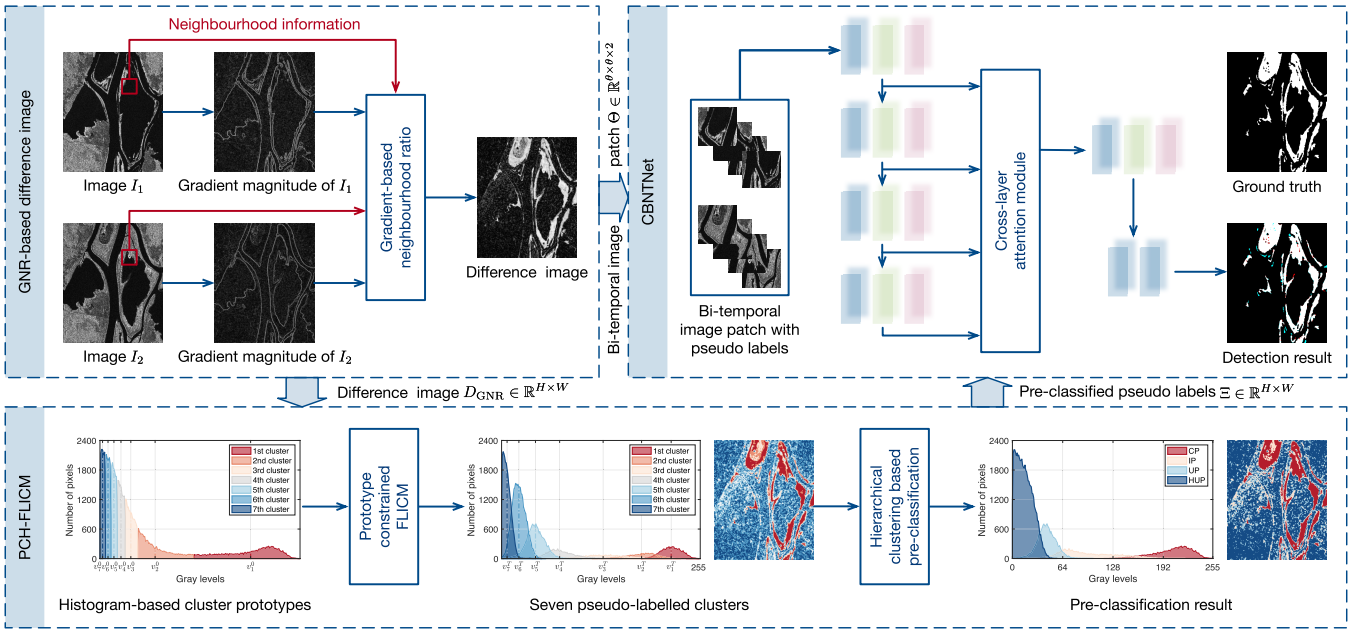

Fig. 1. Flowchart of the proposed EIPCNet. Three key steps are (a) GNR for difference image generation, (b) PCH-FLICM-based preclassification, and (c) CBNTNet for final change detection.

and reliability of preclassification and select high-confidence samples for subsequent network training.

3) Stemming from the attention mechanism, the CBNTNet is constructed to further handle the sample imbalance issue, elevate tolerance to noise, as well as improve change detection accuracy.

The remainder of this article is organized as follows. Section II provides a detailed description of the proposed change detection framework. Experimental datasets and results are presented in Section III. Section IV presents ablation studies, together with parameter analysis on several aspects. Conclusions are drawn in Section V.

II. Methodology

A. Gradient-Based Neighborhood Ratio

Considering bitemporal SAR images I_1 and I_2 captured on the same geographical area at different times, NR operator generates the difference image D_{NR} using

$$D_{\mathrm{NR}}(p) = \alpha_p \times \frac{\min\{I_1(p), I_2(p)\}}{\max\{I_1(p), I_2(p)\}} + (1-\alpha_p) \times \frac{\sum_{q\in\Omega_p \wedge q\neq p} \min\{I_1(q), I_2(q)\}}{\sum_{q\in\Omega_p \wedge q\neq p} \max\{I_1(q), I_2(q)\}} \quad (1)$$

where $I_1, I_2 \in \mathbb{R}^{H\times W}$, $\alpha_p = \sigma(p)/\mu(p)$ denotes the heterogeneity measurement of neighborhood area Ω_p at position p, $\sigma(p)$ and $\mu(p)$ represent the standard deviation and mean of pixel values in Ω_p, respectively. The larger the D_{NR}, the higher similarity between the bitemporal SAR image pair. High α values indicate heterogeneous local areas and low values correspond to homogeneous neighborhoods. The use of local heterogeneity helps restraining speckle noise as well as preserving detailed difference information.

Compared to local heterogeneity characterized by the standard deviation, gradient magnitude can well retain the edge information of the changed regions, so as to better identify all potential changes. To effectively suppress the speckle noise without loss of edge details, we construct the GNR by

$$D_{\mathrm{GNR}}(p) = \left| \frac{\Gamma(p) - \Upsilon(p)}{\Gamma(p) + \Upsilon(p)} \right| \quad (2)$$

$$\Gamma(p) = G_1(p) \times I_1(p) + (1-G_1(p)) \times \frac{\sum_{q\in\Omega_p \wedge q\neq p} I_1(q)}{N-1} \quad (3)$$

$$\Upsilon(p) = G_2(p) \times I_2(p) + (1-G_2(p)) \frac{\sum_{q\in\Omega_p \wedge q\neq p} I_2(q)}{N-1} \quad (4)$$

where N is the total number of pixels in Ω_p, $G_1(p)$ and $G_2(p)$ are the normalized gradient magnitudes of I_1 and I_2 defined as the root mean square of image directional gradients along the horizontal and vertical axes

$$G_1(p) = \sqrt{(I_1(p) \otimes G_r)^2 + (I_1(p) \otimes G_s)^2} \quad (5)$$

$$G_2(p) = \sqrt{(I_2(p) \otimes G_r)^2 + (I_2(p) \otimes G_s)^2} \quad (6)$$

$$G_r = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix}, \; G_s = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix} \quad (7)$$

where $\otimes$ denotes the convolution operation. $G_1(p)$ and $G_2(p)$ are further normalized to obtain values within a dynamic range of [0, 1].

For pixel-based SAR image change detection, it has been proven that compared to the subtraction operator, the ratio operator shows stronger robustness to calibration and radiometric errors [41]. From (2)–(4), we can observe that the proposed GNR characterizes gradient magnitude to balance between intensity level and spatial information of neighboring pixels. The utilization of gradient information further

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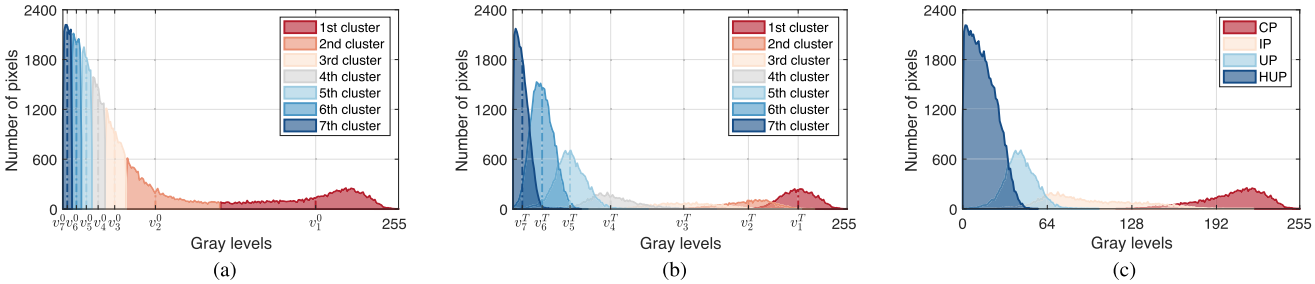

Fig. 2. Step-wise histogram illustrations of PCH-FLICM. (a) Histogram-based cluster prototypes. (b) Prototype constrained FLICM. (c) Grouping-based preclassification.

Algorithm 1 Procedure of PCH-FLICM

Input: The difference image $D_{\mathrm{GNR}} \in \mathbb{R}^{H\times W}$.
Output: Preclassification result $\Xi \in \mathbb{R}^{H\times W}$.

Obtain the upper limit of number of CPs N_{C} via two-class FLICM.
Initialize seven cluster prototypes $V^0 = \{v_1^0, v_2^0, \ldots, v_7^0\}$ according to the histogram distribution of D_{GNR}.
Randomly initialize fuzzy partition matrix.
Set the loop counter $t = 0$, the stopping threshold ϵ, the max number of iterations T, and the control parameter $B = \{\beta_1, \beta_2, \ldots, \beta_7\}$.
for $t \leftarrow 1$ to T **do**
    Update the membership matrix U^t using Eq. 10.
    Calculate the new cluster prototypes V^t using Eq. 11.
    Calculate the fuzzy factor G^t using Eq. 9.
    Calculate the new objective function J_m^t using Eq. 8.
    if $\|J_m^t - J_m^{t-1}\| < \epsilon$ **then**
        break
    else
        $J_m^t \leftarrow J_m^{t-1}$
    end if
end for
Obtain pseudolabeled CPs using Eq. 12.
Obtain pseudolabeled IPs using Eq. 13.
Obtain pseudolabeled UPs using Eq. 14.
Obtain pseudolabeled HUPs.

strengthens the preservation of image details, while weakening the impact of residual multiplicative speckle noise.

B. Prototype-Constrained Hierarchical Fuzzy Local Information C-Means

In addition to coherent speckle noise and loss of edge information, imbalanced class distribution is another issue need to be addressed in SAR change detection. In this article, we propose the PCH-FLICM algorithm that takes into consideration the distribution of pixel intensities between changed and unchanged classes, and also incorporates neighborhood information into hierarchical segmentation of the difference image. The proposed PCH-FLICM provides valuable insights for accurately identifying potential changed regions, as well as finding preliminary and reliable samples for subsequent discrimination. Details of the PCH-FLICM is described as follows with procedures summarized in Algorithm 1 and step-wise histogram illustrations in Fig. 2.

1) Histogram-Based Cluster Prototypes: To avoid incorrect shifts of cluster centers due to imbalanced classes, the preliminary cluster prototypes are initialized based on the histogram distribution of the difference image D_{GNR}. The two-class FLICM algorithm [42] is utilized to divide the difference map into two clusters: changed ϕ_C and unchanged ϕ_U. N_C represents the total number of pixels in class ϕ_C. We select the top N_C samples from D_{GNR} with prominent high intensities and use their mean intensity as cluster prototype of the changed class v_1^0. According to the histogram distribution as shown in Fig. 2, the rest samples are partitioned into six clusters so that each histogram bin contains the same number of pixels. Those cluster prototypes are then determined as the mean pixel intensity of each cluster. It is noted that that the seven cluster prototypes are sorted in descending order $V^0 = \{v_1^0, v_2^0, \ldots, v_7^0\}$ to be used as constraints, preventing inappropriate shifts of clusters $\Phi^0 = \{\phi_1^0, \phi_2^0, \ldots, \phi_7^0\}$.

2) Prototype Constrained Fuzzy Local Information C-Means Clustering: Inspired by [37], the objective function of the proposed prototype-constrained FLICM is defined as

$$J_m = \sum_{i=1}^{HW}\sum_{k=1}^{C}\left[u_{ki}^m\|(1-\beta_k)x_i + \beta_k v_k^0 - v_k\|^2 + G_{ki}\right] \quad (8)$$

where x_i is the gray level value of pixel i, $m = 2$ is the weighting parameter of each fuzzy membership, H and W, respectively, denote the height and width of D_{GNR}, C is the desired number of classes, u_{ki} is the degree of membership of ith pixel in kth cluster, v_k denotes the prototype of kth cluster, β_k represents the control parameter for kth cluster, $\|\cdot\|$ represents the Euclidean distance, and the fuzzy factor G_{ki} is defined as the weighted sum of distances of the cluster prototype from its spatial neighbors by

$$G_{ki} = \sum_{j\in\Omega_i, i\neq j}\frac{1}{d_{ij}+1}(1-u_{kj})^m\|(1-\beta_k)x_j + \beta_k v_k^0 - v_k\|^2 \quad (9)$$

where Ω_i is the neighborhood of pixel i, d_{ij} denotes the spatial distance between pixels i and j, and the element u_{ki} of the membership partition matrix U and cluster prototype v_k can be derived by

$$u_{ki} = \frac{1}{\sum_{j=1}^{C}\left(\dfrac{\|(1-\beta_k)x_i + \beta_k v_k^0 - v_k\|^2 + G_{ki}}{\|(1-\beta_j)x_i + \beta_j v_j^0 - v_j\|^2 + G_{ji}}\right)^{\frac{1}{m-1}}} \quad (10)$$

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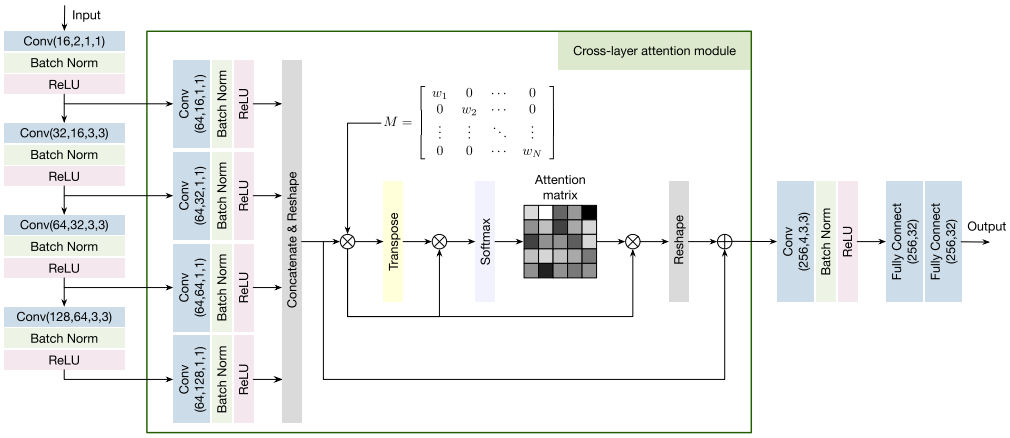

Fig. 3. Network architecture of CBNTNet.

$$v_k = (1-\beta_k)\frac{\sum_{i=1}^{N} u_{ki}^m x_i}{\sum_{i=1}^{N} u_{ki}^m} + \beta_k v_k^0. \tag{11}$$

The iteration stops when the update of membership matrix satisfies the threshold, outputting seven pseudolabeled clusters.

3) Grouping-Based Preclassification: In SAR change detection, unchanged areas often occupy most of the image, resulting in huge number of UPs and extremely imbalanced sample class. The incorporation of spatial information in fuzzy clustering helps preserve insensitiveness to speckle noise and outliers, but nevertheless results in blurred boundaries (where most of the misclassification occurs) between changed and unchanged regions [42]. In this case, we propose to use hierarchical clustering to obtain enough representative samples by grouping seven pseudolabeled clusters into four categories: high-confidence UPs (HUPs), UPs, intermediate pixels (IPs), and CPs. The number of CPs is

$$N_{\mathrm{CP}} = \sum_{i=1}^{\xi_{\mathrm{CP}}} N_i$$
$$\xi_{\mathrm{CP}} = \arg\max_{\xi_{\mathrm{CP}}} \sum_{i=1}^{\xi_{\mathrm{CP}}} N_i \leqslant N_C \tag{12}$$

where the number of clusters in CPs is ξ_{CP}, and N_C denotes the upper limit of N_{CP}. The number of IPs is

$$N_{\mathrm{IP}} = \sum_{j=1}^{\xi_{\mathrm{IP}}} N_{\xi_{\mathrm{CP}}+j}$$
$$\xi_{\mathrm{IP}} = \arg\max_{\xi_{\mathrm{IP}}} N_{\mathrm{CP}} < \sum_{j=1}^{\xi_{\mathrm{IP}}} N_{\xi_{\mathrm{CP}}+j} \leqslant N_{\mathrm{CP}} \times \rho \tag{13}$$

where the number of clusters in IPs is ξ_{IP}, and ρ is the upper limit ratio between N_{IP} and N_{CP}. The number of UPs is

$$N_{\mathrm{UP}} = \sum_{k=1}^{\xi_{\mathrm{UP}}} N_{\xi_{\mathrm{CP}}+\xi_{\mathrm{IP}}+k}$$
$$\xi_{\mathrm{UP}} = \arg\min_{\xi_{\mathrm{UP}}} \sum_{k=1}^{\xi_{\mathrm{UP}}} N_{\xi_{\mathrm{CP}}+\xi_{\mathrm{IP}}+k} > N_{\mathrm{CP}} \tag{14}$$

where the number of clusters in UPs is ξ_{UP}. If $\xi_{\mathrm{CP}} + \xi_{\mathrm{IP}} + \xi_{\mathrm{UP}} < 7$, all pixels that have not been assigned labels will be attributed as HUPs; if $\xi_{\mathrm{CP}} + \xi_{\mathrm{IP}} + \xi_{\mathrm{UP}} < 7$, HUPs will be merged into UPs.

C. Class-Balanced Noise-Tolerant Change Detection Network

On the premise of preclassified pixels with pseudolabels $\Xi \in \mathbb{R}^{H\times W}$, we extract dual-channel patches $\Theta \in \mathbb{R}^{\theta\times\theta\times 2}$ from the bitemporal images through a sliding window of size of $\theta \times \theta$, and assign the labels according to the class of the center pixels. Patches centered at HUPs are pruned from the network training to narrow down the large gap between numbers of samples in changed and unchanged class. To minimize errors involved in these pseudolabels, patches centered at IPs are also discarded. Though only patches with high probabilities are taken as valid samples, the class imbalance and noisy label problems are still inevitable. Therefore, the CBNTNet is constructed for identifying all potential changed regions.

1) Cross-Layer Covariance Attention-Based Network: The attention mechanism [43] imitates the cognitive attention in human vision system and helps the model to flexibly focus on the most relevant parts of the feature map and ignore unimportant information. Existing attention-enhanced networks rarely take account of cross-layer correlations that flow through the convolutional layers. To effectively fuse low-level spatial and high-level semantic information, we insert the cross-layer covariance attention module [44] into the backbone network. The feature maps of multiple layers F are reshaped and weighted by a diagonal matrix M, the attention matrix A is computed as the correlation of M

$$A = \mathrm{softmax}(MF \otimes F^T M^T) \tag{15}$$

and the output is the

$$Z = \psi(A \otimes MF) + MF \tag{16}$$

where ψ denotes the reshape function. The detailed network structure is illustrated in Fig. 3.

2) Class-Balanced and Noise-Tolerant Loss: The binary focal loss (BFL) [45], originally proposed for object detection, is a useful tool to tackle the class imbalance problem through introducing a balancing factor and a modulating term to

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the binary cross entropy loss (BCEL). The class-balanced adaptive focal loss (CBAFL) [24] set the balancing factor that adaptively adjusts the weight of minority and majority classes. The CBAFL is defined as

$$\mathcal{L}_{\text{CBAFL}} = \begin{cases} -\sigma(1-\hat{y})^{\gamma}\log(\hat{y}) & y=1 \\ -(1-\sigma)\hat{y}^{\gamma}\log(1-\hat{y}) & y=0 \end{cases} \tag{17}$$

where y and $\hat{y}$ are the class label and the output probability of the network, respectively. $\sigma = (N_{\text{CP}}/N_{\text{UP}} + N_{\text{CP}})$ denotes the balancing factor, and γ denotes the focusing parameter to alleviate the influence of samples that are easy to be classified in the training and force the model to pay more attention to those difficult to train. Though the proposed PCH-FLICM algorithm generates reliable pseudolabels, it is unavoidable to involve some errors that would hamper the model training. The mean absolute error loss (MAEL) has been empirically demonstrated to be robust to noisy labels calculated by

$$\mathcal{L}_{\text{MAEL}} = \|\hat{y} - y\|. \tag{18}$$

To combine the respective strength of CBAFL and MAEL, we design the class-balanced noise-tolerant loss (CBNTL) as

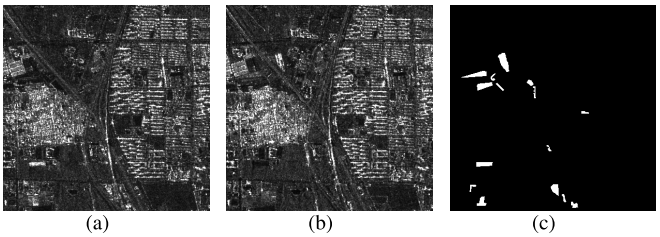
$$\mathcal{L}_{\text{CBNTL}} = \lambda\mathcal{L}_{\text{CBAFL}} + (1-\lambda)\mathcal{L}_{\text{MAEL}} \tag{19}$$

where λ is the hyperparameter controlling the trade-off between class imbalance and label noise.

III. Experimental Datasets and Results

A. Dataset Description

In order to verify the effectiveness of the proposed SAR change detection approach in various scenarios, experiments were carried out on five datasets. The Daxing-I and Daxing-II datasets were captured by Gaofen-3 satellite in Daxing, Beijing. Bitemporal image pairs were acquired in April 2017 and May 2018, respectively. Similar image acquisition seasons can reduce the impact of vegetation growth on backscattering, minimizing false alarms caused by vegetation changes. As illustrated in Fig. 4, images of Daxing-I dataset are of size of 400 × 400 pixels, and it is obvious that the proportion of CPs is much smaller compared with UPs. The main causes of backscatter changes in this area are the construction and demolition of buildings. Due to the update of urban construction, area of the Daxing-II dataset experienced significant changes. Its image size is 215 × 248 shown in Fig. 5. Figs. 6 and 7 present the Coast and Yellow River datasets, in which images were acquired by Radarsat-2 satellite at the Yellow River Delta region of China in June 2008 and June 2009, respectively. Backscatter changes occurred in the water-side areas due to variations in water bodies. The image sizes are 280 × 450 and 444 × 291 pixels, respectively. The Ottawa dataset in Fig. 8 consists of Radarsat-1 images of sizes of 350 × 290 pixels taken from the Ottawa region of Canada during the summer of 1997. The recession of floods during that period was the main cause for backscatter changes in the SAR images. From the reference images in Figs. 4 and 6, it can be visually observed that the proportion of CPs in the Daxing-I and Coast datasets is much smaller compared with UPs. On the contrary, the Daxing-I, Yellow River and Ottawa datasets are relatively balanced with class distributions. Details of five datasets are summarized in Table I.

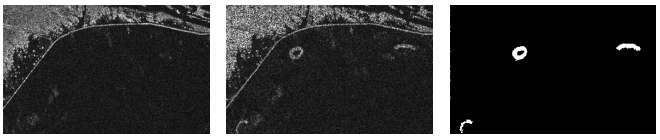


(a) (b) (c)

Fig. 4. Daxing-I dataset (a) image acquired in April 2017, (b) image acquired in May 2018, and (c) reference image.

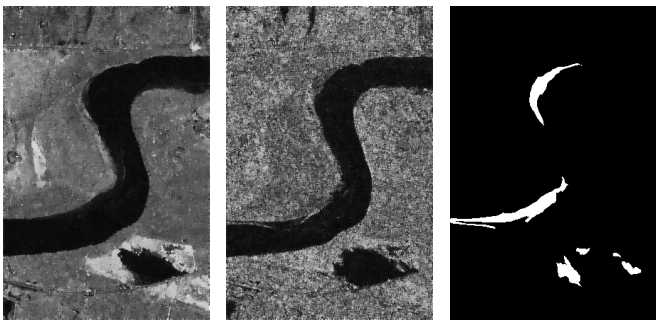
(a) (b) (c)

Fig. 5. Daxing-II dataset (a) image acquired in April 2017, (b) image acquired in May 2018, and (c) reference image.



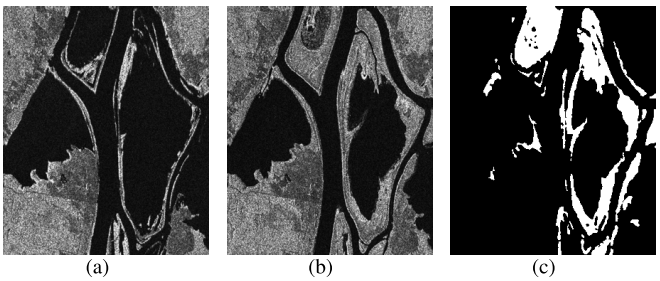
(a) (b) (c)

Fig. 6. Coast dataset (a) image acquired in June 2008, (b) image acquired in June 2009, and (c) reference image.



(a) (b) (c)

Fig. 7. Yellow River dataset (a) image acquired in June 2008, (b) image acquired in June 2009, and (c) reference image.



(a) (b) (c)

Fig. 8. Ottawa dataset (a) image acquired in May 1997, (b) image acquired in August 1997, and (c) reference image.

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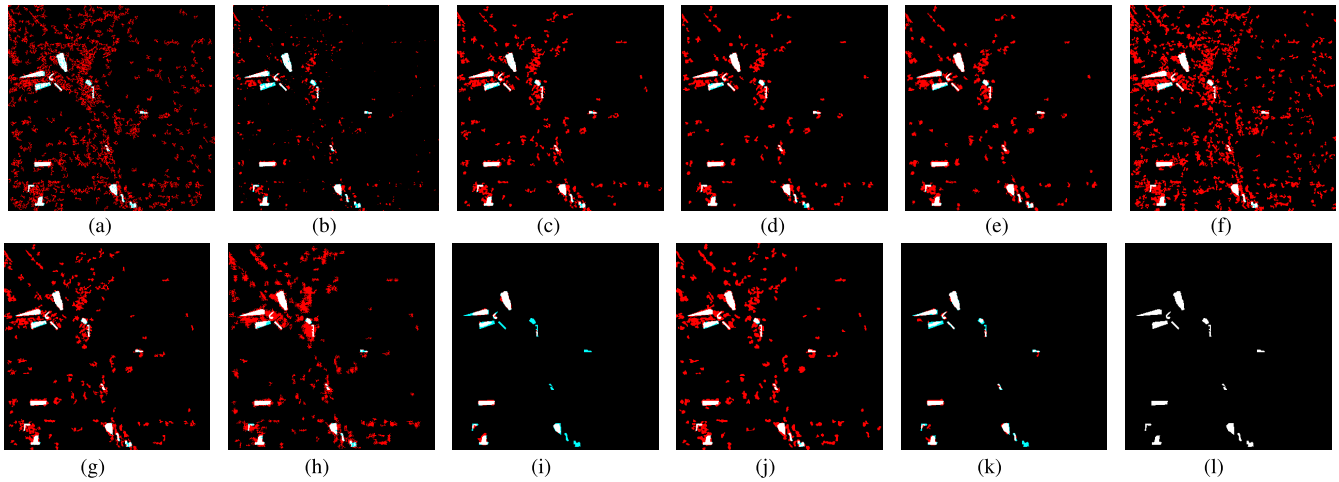

Fig. 9. Visualization of various change detection methods on Daxing-I dataset (a) LR-FCM, (b) LR-FLICM, (c) PCA-KM, (d) NR-ELM, (e) GaborPCANet, (f) CWNN, (g) DDNet, (h) SAFNet, (i) RUSACD, (j) LANTNET, (k) proposed, and (l) ground truth.

TABLE I
SUMMARY OF FIVE EXPERIMENTAL DATASETS

Dataset	Size	N_{CP}	N_{UP}	$N_{\mathrm{CP}} : N_{\mathrm{UP}}$
Daxing-I	400×400	2,418	157,582	1:65
Daxing-II	215×248	6,494	46,826	1:7
Coast	280×450	1,348	124,652	1:92
Yellow River	444×291	4,255	124,949	1:29
Ottawa	350×290	16,049	85,451	1:5

B. Evaluation Criteria

Performance of the proposed method was evaluated based on five commonly adopted metrics: false positive (FP) refers to the number of UPs that are incorrectly detected as changed ones; false negative (FN) denotes the number of CPs misclassified as unchanged; overall error (OE) indicates the OE as the sum of FP and FN; percentage correct classification (PCC) represents the percentage of correctly classified pixels out of the total amount using

$$\mathrm{PCC} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN}} \tag{20}$$

where TP and TN refer to the true positive and true negative, respectively; Kappa coefficient (KC) is the statistical measure to assess the performance of the algorithm in terms of consistency and accuracy calculated by

$$\mathrm{KC} = \frac{\mathrm{PCC} - \mathrm{PRE}}{1 - \mathrm{PRE}} \tag{21}$$

$$\mathrm{PRE} = \frac{(\mathrm{TP} + \mathrm{FN}) \times (\mathrm{TP} + \mathrm{FP}) + (\mathrm{TN} + \mathrm{FP}) \times (\mathrm{TN} + \mathrm{FN})}{(\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN})^2}. \tag{22}$$

C. Experimental Setup and Parameter Settings

The GNR-based difference image generation and the PCH-FLICM-based preclassification were run in MATLAB 2020a, and the CBNTNet training was implemented on Google Colab with NVIDIA Tesla T4 GPU with 16-GB memory and PyTorch framework.

Hyperparameters for each step were set separately. In GNR-based difference image generation, the neighourhood Ω_p was set as a 3×3 window centered at position p. In PCH-FLICM, the weighting parameter of each fuzzy membership was chosen as $m = 2$. The iteration stopped when the loop counter reached the maximum number of iterations $T = 200$ or stopping threshold was satisfied $\epsilon = 10^{-5}$. The control parameters for the first cluster was set as $\beta_1 = 0.5$ and others were calculated using $\beta_i = 0.8\beta_{i-1}$. $\rho = 2$ was used as the upper limit ratio between N_{IP} and N_{CP}. Consequently, the difference image were preclassified into four pseudolabeled clusters: HUPs, UPs, Ips, and CPs. For CBNTNet, dual-channel patches were collected from bitemporal images, each of which was labeled according to preclassified pseudolabel of its center pixel. To mitigate class imbalance and noisy label issues, patches centered at HUPs and IPs were discarded. Only patches centered at UPs and CPs were considered as training samples, and the trained network reclassifies all the patches to further cope with errors involved in preclassified pseudolabels. In the network training, all convolutional layers were initialized with weights drawn from Gaussian distributions with zero means and standard deviation of $(2/(K_i \times K_i \times c_{\mathrm{out},i}))^{1/2}$, where K_i is the kernel size and $c_{\mathrm{out},i}$ represents the number of output channels in ith convolutional layer. Weights of the two fully connected layers were sampled from zero-mean Gaussian distributions with a standard deviation of 0.01. Weights of batch normalization layers and biases were primally set as 1 and 0, respectively. The CBNTNet was trained with Adam optimizer and a momentum of 0.9. Each mini-batch contains 128 sample patches. Learning rate was set as 0.001 and number of epochs set to 100. In terms of trade-off between CBAFL and MAEL, the balancing factor between changed and unchanged class was $\sigma = (N_{\mathrm{CP}}/N_{\mathrm{UP}} + N_{\mathrm{CP}})$, the focus parameter was set as $\gamma = 2$, and the weight parameter was $\lambda = 0.7$.

D. Experimental Results and Comparisons

In this section, visual and quantitative results are reported and compared with various state-of-the-art methods, including the FCM algorithm based on LR difference map (LR-FCM), FLICM algorithm based on LR difference map

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TABLE II
CHANGE DETECTION RESULTS OF VARIOUS METHODS ON DAXING-I DATASET

Methods	FP	FN	OE	PCC(%)	KC(%)
LR-FCM	16416	368	16768	89.52	17.47
LR-FLICM	3680	448	4112	97.42	47.79
PCA-KM [46]	8048	192	8240	94.85	33.45
NR-ELM [35]	6208	208	6416	95.99	39.55
GaborPCANet [47]	6800	128	6928	95.67	38.32
CWNN [13]	21152	1072	21264	86.71	15.56
DDNet [48]	8608	154	8762	94.52	32.38
RUSACD [37]	317	861	1178	99.26	72.19
SAFNet [49]	10523	248	10771	93.27	26.86
LANTNET [44]	9818	**98**	9916	93.80	30.12
EIPCNet	**486**	583	**1069**	**99.33**	**77.10**

TABLE III
CHANGE DETECTION RESULTS OF VARIOUS METHODS ON DAXING-II DATASET

Methods	FP	FN	OE	PCC(%)	KC(%)
LR-FCM	469	1343	1812	96.60	83.15
LR-FLICM	261	1242	1503	97.18	85.87
PCA-KM [46]	192	1120	1312	97.54	87.72
NR-ELM [35]	459	538	10027	98.12	91.18
GaborPCANet [47]	501	608	1109	97.92	90.21
CWNN [13]	959	**219**	1178	97.79	90.16
DDNet [48]	619	343	962	98.20	91.72
RUSACD [37]	**106**	1015	1121	97.90	89.54
SAFNet [49]	1054	344	1398	97.38	88.30
LANTNET [44]	906	328	1234	97.69	89.58
EIPCNet	431	407	**838**	**98.43**	**92.67**

TABLE IV
CHANGE DETECTION RESULTS OF VARIOUS METHODS ON COAST DATASET

Methods	FP	FN	OE	PCC(%)	KC(%)
LR-FCM	13978	257	14235	88.70	11.55
LR-FLICM	2590	174	2764	97.81	45.07
PCA-KM [46]	32299	25	32324	74.35	5.63
NR-ELM [35]	5142	21	5163	95.90	32.76
GaborPCANet [47]	23517	**7**	23524	81.33	8.37
CWNN [13]	16191	50	16241	87.11	12.03
DDNet [48]	5663	59	5722	95.46	29.80
RUSACD [37]	130	199	329	99.74	87.34
SAFNet [49]	15236	230	15466	87.73	10.87
LANTNET [44]	8163	85	8248	93.45	21.98
EIPCNet	**91**	167	**258**	**99.80**	**90.05**

(LR-FLICM), principal component analysis (PCA) and K-means algorithm (PCA-KM) [46], extreme learning machine based on neighborhood ratio (NR-ELM) [35], Gabor PCA network (GaborPCANet) [47], convolutional-wavelet neural networks (CWNNs) [13], dual-domain network (DDNet) [48], Siamese adaptive fusion network (SAFNet) [49], robust unsupervised small area change detection method (RUSACD) [37], and layer attention-based noise tolerant network (LANTNET) [44]. In order to highlight the locations of FPs and FNs, the change detection results in this article are presented using black, white, red, and cyan colors. Specifically, correctly detected UPs are represented by black, correctly classified CPs are in white, FPs and FNs are denoted by red and cyan, respectively.

1) Results on Daxing-I Dataset: In Daxing-I dataset, compared with changes related to construction and demolition of buildings, changes occurring between bare land and vegetation are less distinct and more scattered, hindering the accurate detection of all changes. Visual results of various methods are shown in Fig. 9 with corresponding quantitative results presented in Table II. It is noted that with the exception of RUSACD and our EIPCNet, all other methods generate a large number of FPs, particularly LR-FCM and CWNN. The main reason for the high number of FPs in LR-FCM is the lack of use of contextual information, while CWNN is primarily designed for detecting changes in large-scale sea ice. Owing to the complexity of change types within the region, only the RUSACD and proposed methods achieved a KC of 70%. The proposed EIPCNet had also been shown robustness to noise and reached the highest KC value (77.10%), outperforming RUSACD with a relatively high number of FPs. It can also be concluded that EIPCNet exhibited outstanding performance for small area change detection from bitemporal SAR images.

2) Results on Daxing-II Dataset: Compared to the Daxing-I dataset, the Daxing-II images exhibits relatively homogeneous types of changes, mainly caused by transitions between bare land and artificial constructions. Therefore, various methods have shown working well, as shown in Fig 10. Clustering-based methods (such as LR-FCM, LR-FLICM, and PCA-KM) suffered from noise introduced by uneven backscattering in the building area and missed many changed details. Conversely, CWNN, DDNet, SAFNet, and LANTNET generated relatively high FPs. In terms of quantitative results displayed in Table III, nearly half of the methods achieved a KC of over 90%. Among them, RUSACD had the fewest FPs of only 106, while the proposed EIPCNet, though not optimal in terms of FPs and FNs, achieved a more balanced trade-off between them, resulting in the highest KC of 92.67% and showing superior robustness in suppressing speckle noise and retaining details.

3) Results on Coast Dataset: Compared to the complexity of detecting urban area changes in Daxing-I and Daxing-II datasets, Coast and Yellow River datasets primarily focus on locating changes at water-land boundaries. In Coast dataset, noise in land areas had a fundamental influence on the final detection result. From Fig. 11, it is noted that all algorithms are able to roughly identify all changed regions, and their differences mainly lie in the suppression of FPs. As illustrated in Table IV, LR-FCM, PCA-KM, GaborPCANet, CWNN, and SAFNet yielded high FPs of 11.09%, 25.63%, 18.66%, 12.85%, and 12.09%, respectively. It worth mentioning that our proposed algorithm achieved far superior performance (the fewest FPs of 91 and the highest KC of 90.05%) to the state-of-the-art methods, declaring its robustness in detecting all meaningful changes, particular for small area change detection with strong speckle noise.

4) Results on Yellow River Dataset: From Fig. 12, it can be observed that in the Yellow River dataset, pseudochanges in the land areas had a minor impact on the final change detection results. Except for LR-FCM, all algorithms exhibited good noise suppression effects and achieved a KC of over 70%. Among them, GaborPCANet has the fewest FPs pixels with only 646. In comparison, CWNN has the fewest FNs, with only 496 pixels. In line with Table V, the KC value of the proposed algorithm still outperformed CWNN, DDNet,

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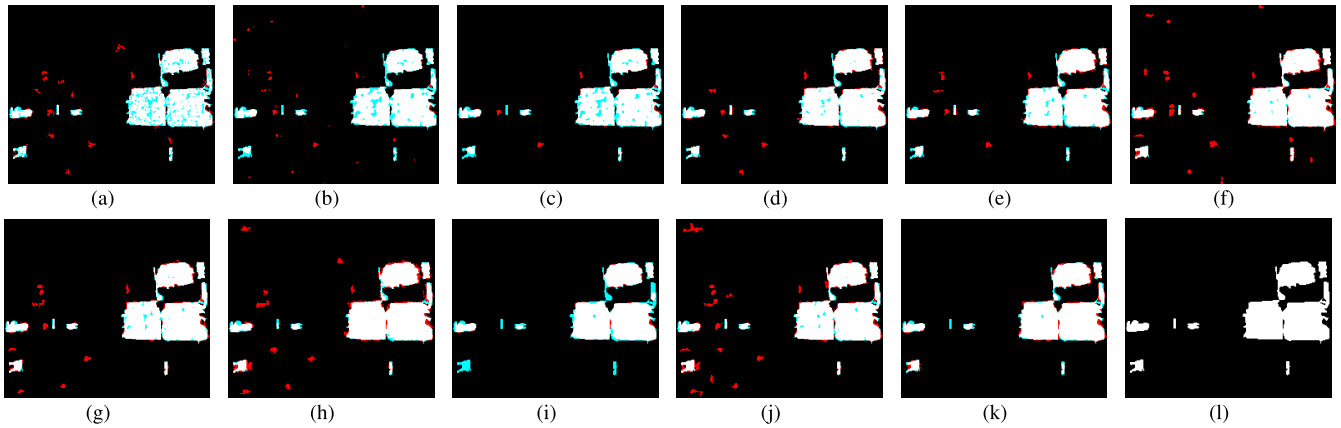

Fig. 10. Visualization of various change detection methods on Daxing-II dataset (a) LR-FCM, (b) LR-FLICM, (c) PCA-KM, (d) NR-ELM, (e) GaborPCANet, (f) CWNN, (g) DDNet, (h) SAFNet, (i) RUSACD, (j) LANTNET, (k) proposed, and (l) ground truth.

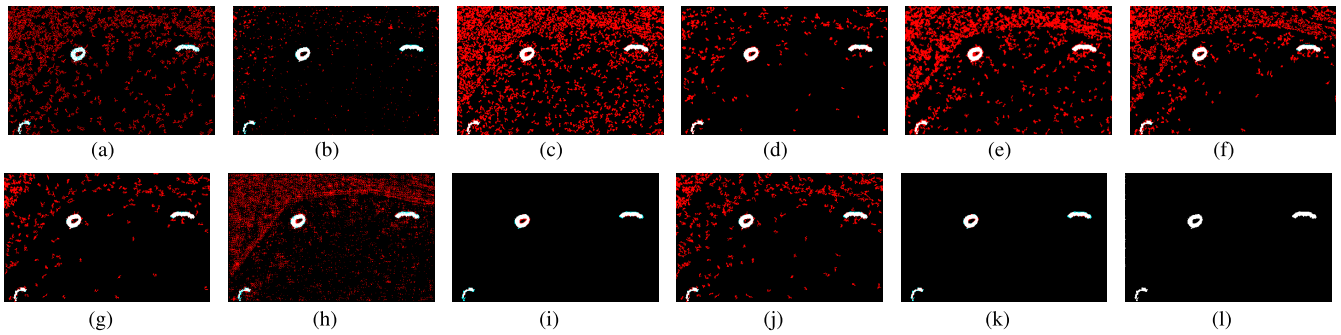

Fig. 11. Visualization of various change detection methods on Coast dataset (a) LR-FCM, (b) LR-FLICM, (c) PCA-KM, (d) NR-ELM, (e) GaborPCANet, (f) CWNN, (g) DDNet, (h) SAFNet, (i) RUSACD, (j) LANTNET, (k) proposed, and (l) ground truth.

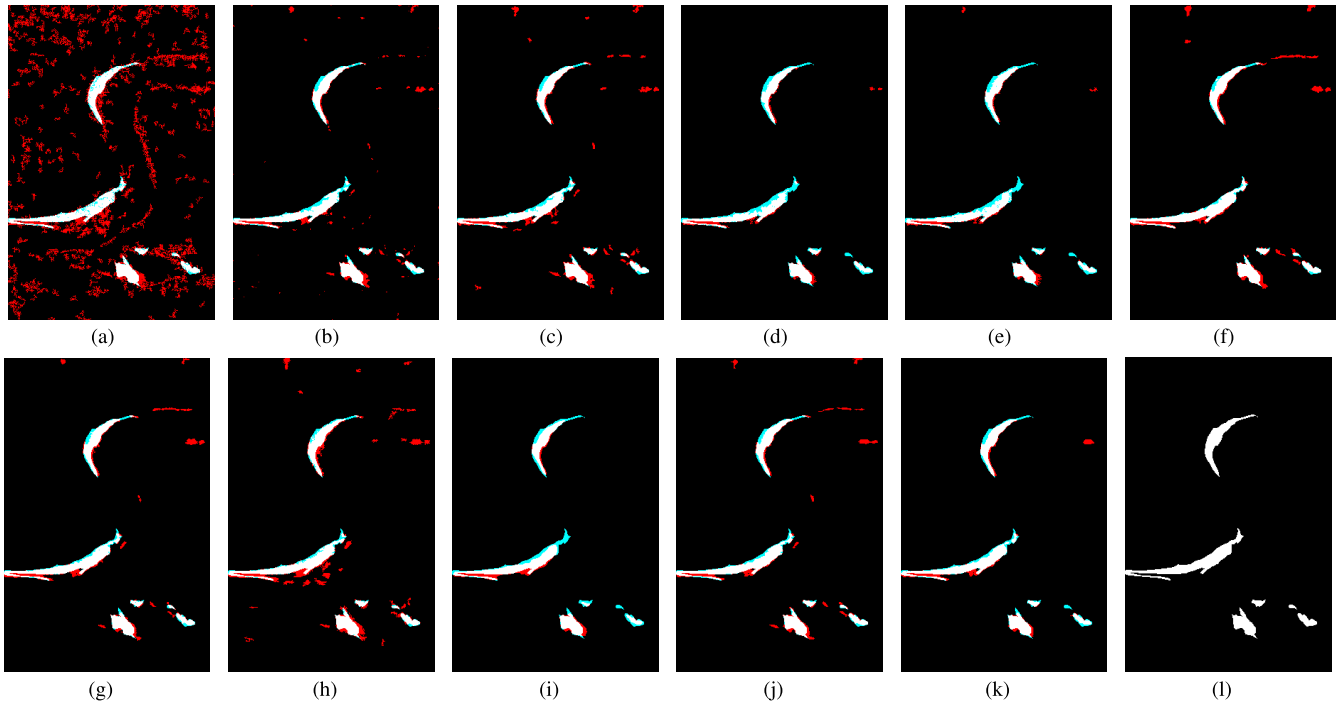

Fig. 12. Visualization of various change detection methods on Yellow River dataset (a) LR-FCM, (b) LR-FLICM, (c) PCA-KM, (d) NR-ELM, (e) GaborPCANet, (f) CWNN, (g) DDNet, (h) SAFNet, (i) RUSACD, (j) LANTNET, (k) proposed, and (l) ground truth.

RUSACD, SAFNet, and LANTNET by 3.16%, 3.53%, 4.91%, 10.24%, and 1.19%, respectively. It is evident that the proposed EIPCNet had the least number of erroneous pixels, totaling 1527, and reached the highest KC of 81.09%,

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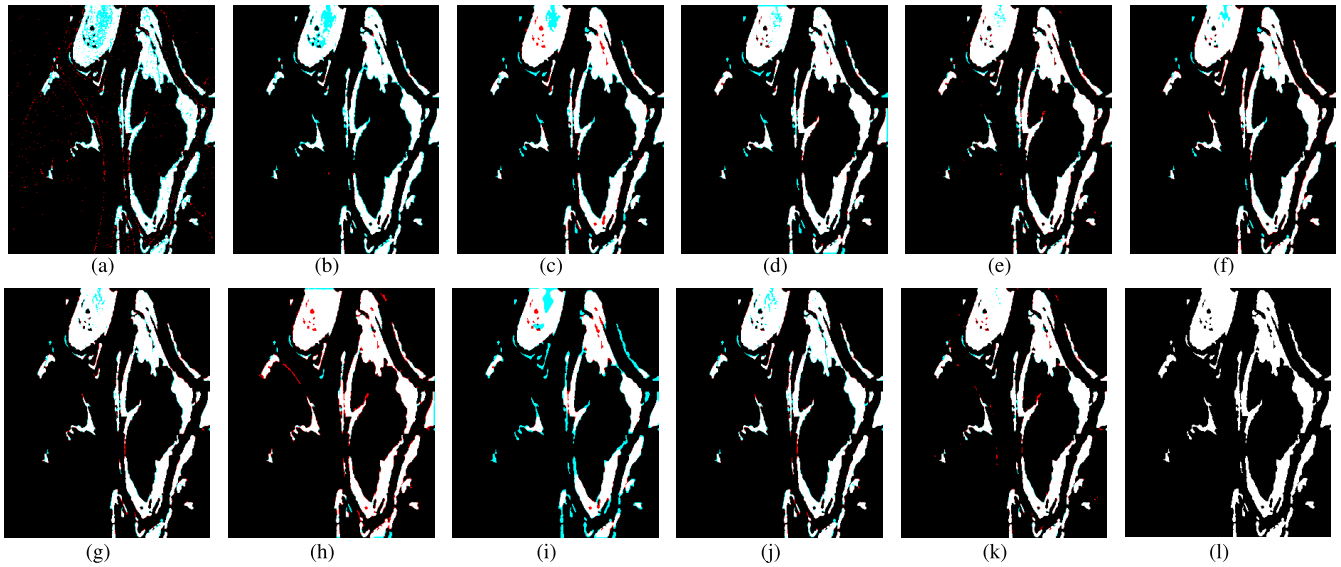

Fig. 13. Visualization of various change detection methods on Ottawa dataset (a) LR-FCM, (b) LR-FLICM, (c) PCA-KM, (d) NR-ELM, (e) GaborPCANet, (f) CWNN, (g) DDNet, (h) SAFNet, (i) RUSACD, (j) LANTNET, (k) proposed, and (l) ground truth.

TABLE V
CHANGE DETECTION RESULTS OF VARIOUS METHODS ON YELLOW RIVER DATASET

Methods	FP	FN	OE	PCC(%)	KC(%)
LR-FCM	12265	517	12782	90.11	33.47
LR-FLICM	1344	788	2132	98.35	75.63
PCA-KM [46]	673	1693	2366	98.17	74.21
NR-ELM [35]	599	1229	1828	98.59	76.08
GaborPCANet [47]	**646**	1219	1865	98.56	75.76
CWNN [13]	1534	**496**	2030	98.43	77.93
DDNet [48]	1344	648	1992	98.46	77.56
RUSACD [37]	650	1190	1840	98.58	76.18
SAFNet [49]	2284	579	2863	97.78	70.85
LANTNET [44]	1032	687	1719	98.67	79.90
EIPCNet	680	847	**1527**	**98.82**	**81.09**

TABLE VI
CHANGE DETECTION RESULTS OF VARIOUS METHODS ON OTTAWA DATASET

Methods	FP	FN	OE	PCC(%)	KC(%)
LR-FCM	1182	3548	4730	95.34	81.38
LR-FLICM	**133**	2685	2818	97.22	88.85
PCA-KM [46]	648	1908	2556	97.48	90.23
NR-ELM [35]	492	1519	2011	98.02	92.36
GaborPCANet [47]	823	926	1749	98.28	93.51
CWNN [13]	754	973	1727	98.30	93.57
DDNet [48]	587	1015	1602	98.42	94.01
RUSACD [37]	536	3443	3979	96.08	84.11
SAFNet [49]	1742	**687**	2429	97.61	91.25
LANTNET [44]	638	911	1549	98.47	94.23
EIPCNet	607	438	**1045**	**98.97**	**96.15**

confirming its usefulness in noise suppression and image detail preservation.

5) Results on Ottawa Dataset: The main reason for image changes in the Ottawa dataset is the alteration in surface scattering caused by the receding floodwaters. Compared to Coast and Yellow River datasets, the changed areas in the Ottawa dataset are more scattered, as shown in Fig. 13. From Table VI, the RUSACD algorithm, which performed considerably well on the other four datasets, generates the highest missed detections of 3443, probably due to the incorporation of errors embedded in pseudolabeled training samples. Most of the missed detections occur on the boundary between changed and unchanged areas. On the contrary, SAFNet generates the highest FPs of 1742, mainly along the edges. Our proposed EIPCNet could make full use of edge information to alleviate noise disturbance, hence produced the least number of false detection, indicating the strongest noise resilient capability.

IV. ABLATION STUDIES AND ANALYSIS

In this section, performance of the proposed algorithm will be analyzed in the following aspects: 1) analysis of difference map generation methods; 2) analysis of sample imbalance; 3) comparative analysis of preclassification methods; 4) impact of input patch size; 5) analysis of varying weight parameters; and 6) analysis of different types of loss functions.

A. Analysis of Difference Map Generation Methods

The GNR-based difference maps of five different datasets are shown in Fig. 14(a). It can be seen that D_{GNR} accurately demonstrated difference information between the bitemporal SAR image pair. Two common evaluation metrics were used to assess and compare the performance of the difference maps: receiver operating characteristic (ROC) curve and area under the ROC curve (AUC) [20], [21], [36]. The ROC curve plots the false alarm rate (horizontal axis) against the detection rate (vertical axis). The closer the curve is to the upper left corner or the higher it is, the higher the detection rate is at the same false alarm rate, indicating stronger description of difference information. From Fig. 15, the GNR curve stayed on the top of LR and NR. According to Table VII, without incorporation of neighborhood information, the AUC values of LR were much lower than other methods. The AUC value of GNR was only slightly outperformed by the LMR on Ottawa dataset, showing its stability and superiority in suppressing noise and retaining edge information. In Fig. 16, the grayscale levels of the

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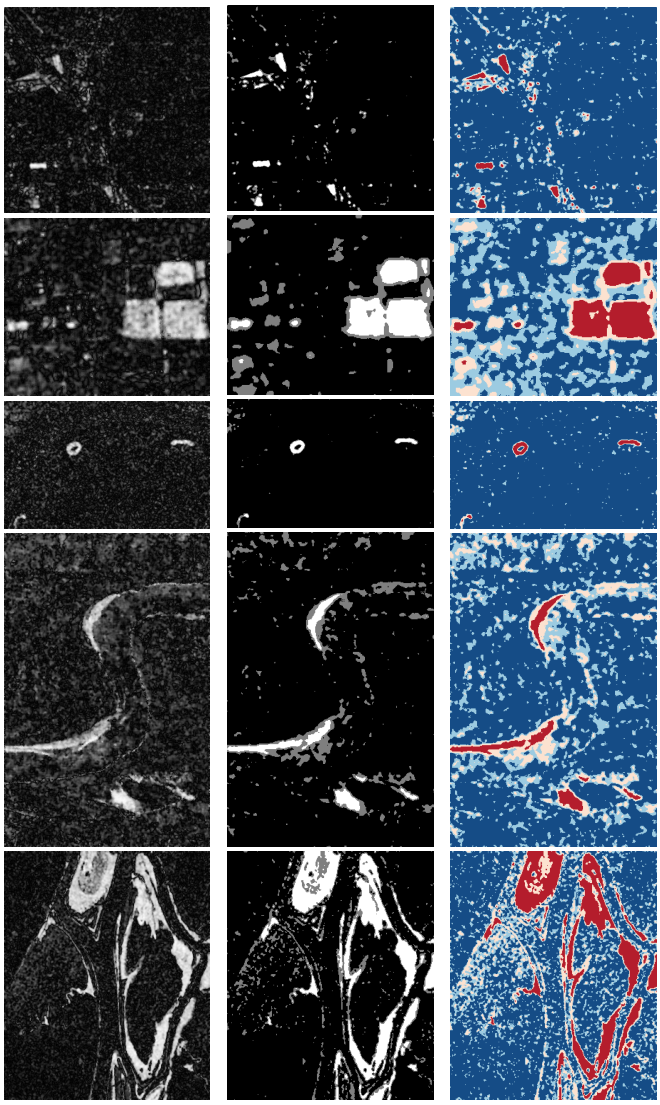

Fig. 14. Step-wise results of the proposed method on Daxing-I, Daxing-II, Coast, Yellow River, and Ottawa datasets, (a) GNR-based difference maps. (b) Preclassification results after merging HUPs into Ups. (c) Preclassified HUPs (in Yale blue), UPs (in pale cerulean), IPs (in lumber), and CPs (in upsdell red).

TABLE VII

COMPARISONS OF AUC VALUES WITH DIFFERENT TYPES OF DIFFERENCE IMAGE GENERATION METHODS

Dataset	AUC			
	LR	LMR	NR	GNR
Daxing-I	0.8821	0.9624	0.9674	**0.9723**
Daxing-II	0.9255	0.9850	0.9859	**0.9922**
Coast	0.8508	0.9731	0.9833	**0.9925**
Yellow River	0.9162	0.9742	0.9794	**0.9801**
Ottawa	0.9754	**0.9970**	0.9940	0.9958

pre- and posttemporal images and the difference map are used as the x-, y-, and z-axes to construct a 3-D space. It is evident that different categories of pixels exhibit distinct layering characteristics, further demonstrating its discriminative ability of the proposed GNR-based difference map.

B. Analysis of Sample Imbalance

In SAR change detection, UPs often occupy the majority of the image, resulting in a much larger number of unchanged samples selected through unsupervised methods compared to changed ones. This is mainly owing to the presence of a large number of HUPs that are easily determined and leads to imbalanced sample quantities. To mitigate the impact of sample imbalance on network model training, effective measures need to be taken. Fig. 14(b) shows the three-class preclassification results by merging HUPs into UPs, where black, gray, and white represent UPs, Ips, and CPs. It is obvious that the number of UPs were much larger than those of CPs, causing a disproportionate ratio between unchanged and changed samples that could interfere with subsequent network training. As illustrated in Fig. 14(c), the proposed PCH-FLICM clustering distinguished the samples into four categories: HUPs (in Yale blue), UPs (in pale cerulean), IPs (in lumber), and CPs (in upsdell red). It can be observed that the utilization of hierarchical clustering algorithm with prototype constraints and the removal of HUPs from the network training helped alleviate sample imbalance to a great extent.

C. Comparative Analysis of Preclassification Methods

Preclassification is crucial step for selecting and constructing high-confidence changed and unchanged samples through unsupervised methods. The quality of the preclassification results directly affects the quality of the samples and ultimately impacts the change detection results. To confirm the usefulness of the proposed GNR and PCH-FLICM, we chose the Coast dataset as an example and compared its performance with several commonly used preclassification methods for SAR image change detection, including the cross combinations of difference map generation methods (LR, LMR, NR, and GNR) and clustering algorithms (three-class FCM, three-class FLICM, and hierarchical FCM). For the convenience of result comparison and display, the HUPs were merged into UPs in the preclassification results. Preclassification results of various methods are shown in Fig. 17. Parameters UU, UI, UC, CU, CI, and CC were calculated, where the first letter represents the pixel category in the ground truth, and the second letter denotes the pixel category in the preclassification result. Preclassification results were then assessed using four metrics: initial correct volume (ICV), initial erroneous volume (IEV), percentage of samples (PS), and sample accuracy (SA) calculated by

$$\mathrm{ICV} = \frac{\mathrm{CC}+\mathrm{UU}}{\mathrm{UU}+\mathrm{UI}+\mathrm{UC}+\mathrm{CU}+\mathrm{CI}+\mathrm{CC}} \tag{23}$$

$$\mathrm{IEV} = \frac{\mathrm{CU}+\mathrm{UC}}{\mathrm{UU}+\mathrm{UI}+\mathrm{UC}+\mathrm{CU}+\mathrm{CI}+\mathrm{CC}} \tag{24}$$

$$\mathrm{PS} = \mathrm{ICV}+\mathrm{IEV} \tag{25}$$

$$\mathrm{SA} = \frac{\mathrm{CC}+\mathrm{UU}}{\mathrm{UU}+\mathrm{UC}+\mathrm{CU}+\mathrm{CC}}. \tag{26}$$

It can be seen from Table VIII that in terms of sample quantity, the proposed GNR-PCH-FLICM correctly selected the highest number of samples (99.03%) among all the preclassification methods. In terms of SA, only a total of 90 erroneous labels

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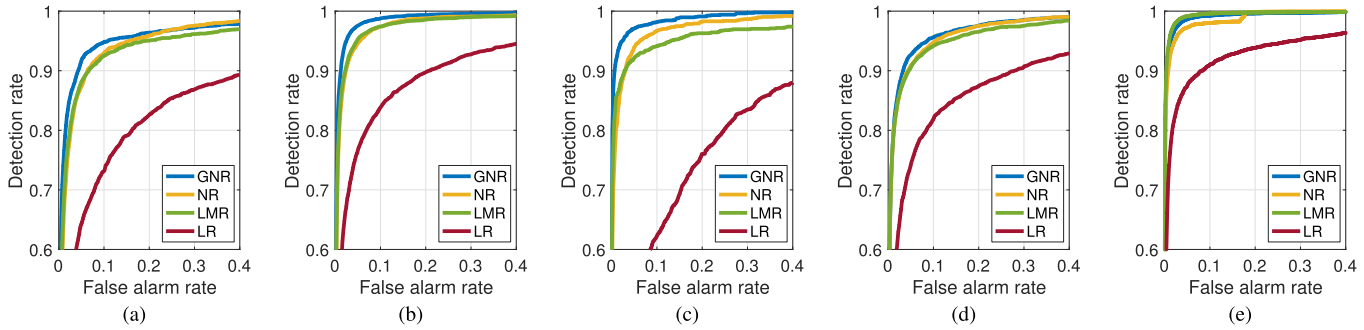

Fig. 15. ROC curves of the difference images obtained using various methods on (a) Daxing-I dataset, (b) Daxing-II dataset, (c) Coast dataset, (d) Yellow River dataset, and (e) Ottawa dataset.

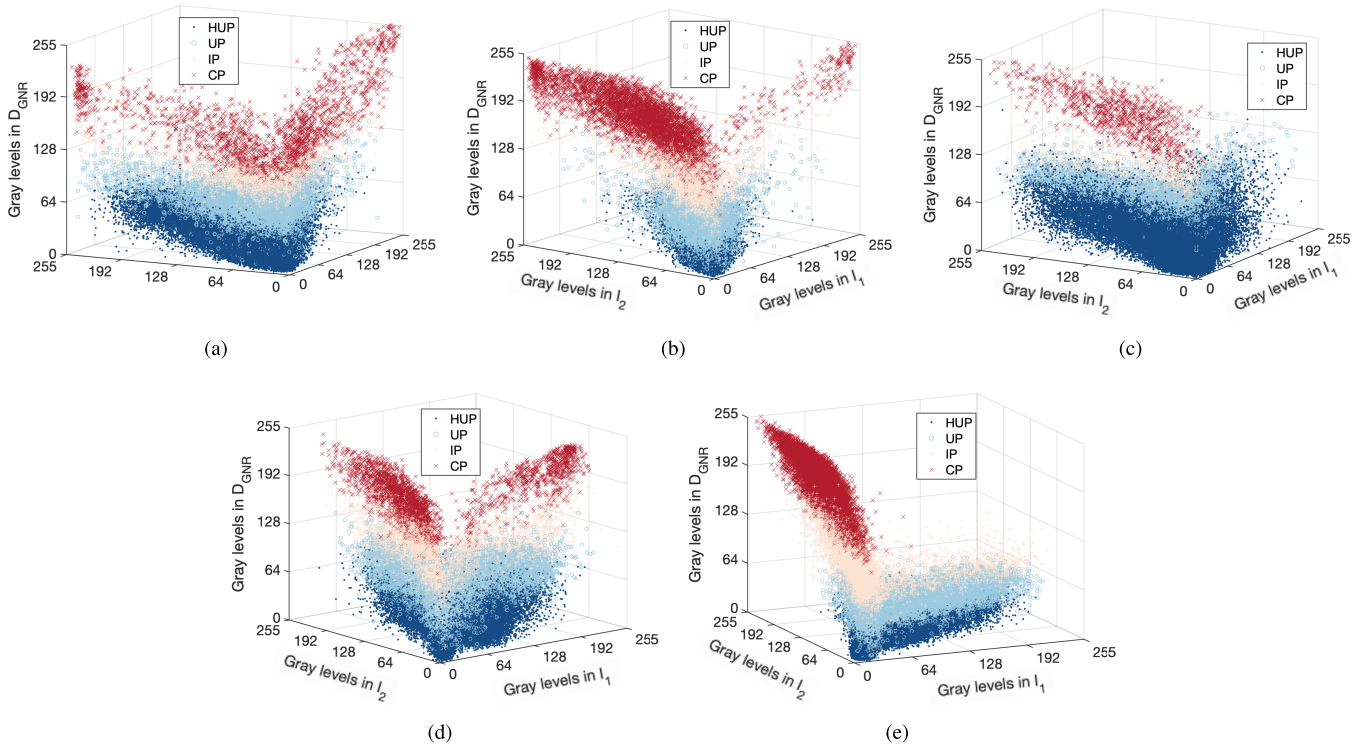

Fig. 16. Three-dimensional visualization of preclassification, where x-, y-, and z-axis represents the grayscale levels of the pre- and posttemporal SAR images and the difference map (a) Daxing-I dataset, (b) Daxing-II dataset, (c) Coast dataset, (d) Yellow River dataset, and (e) Ottawa dataset.

TABLE VIII

PRECLASSIFICATION RESULTS ON COAST DATASET

Methods	UU	UI	UC	CU	CI	CC	ICV	IEV	PS	SA
LR-FCM3	92541	9184	22927	257	91	1000	74.24	18.40	92.64	80.14
LMR-FCM3	96436	26681	1525	48	140	1160	77.46	1.26	78.71	98.40
NR-FCM3	143155	10591	10906	340	256	752	82.47	8.93	91.39	90.23
GNR-FCM3	123378	1147	127	107	193	1048	98.75	0.19	98.94	99.81
LR-HFCM	109158	10452	5042	465	267	616	87.12	4.37	91.49	95.22
LMR-HFCM	110192	10988	3472	74	62	1212	88.42	2.81	91.23	96.92
NR-HFCM	115947	4710	3995	687	269	392	92.33	3.72	96.05	96.13
GNR-HFCM	112359	9523	2770	25	44	1279	90.19	2.22	92.41	97.60
LR-FLICM3	84964	26314	13374	32	52	1264	68.43	10.64	79.07	86.54
LMR-FLICM3	100117	23911	624	33	107	1208	80.42	0.52	80.94	99.36
NR-FLICM3	104902	14810	4940	276	381	691	83.80	4.14	87.94	95.29
GNR-FLICM3	100456	23994	202	5	151	1192	80.67	0.16	80.84	99.80
LR-PCH-FLICM	121396	2652	604	391	294	663	96.87	0.79	97.66	99.19
LMR-PCH-FLICM	121760	2767	125	99	190	1059	97.48	0.18	97.65	99.82
NR-PCH-FLICM	112829	7780	4043	513	225	610	90.03	3.62	93.65	96.14
GNR-PCH-FLICM	123778	861	13	77	355	916	**98.96**	**0.07**	**99.03**	**99.93**

were involved, accounting for only 0.07% of the total number of pixels, which is the highest accuracy among all methods. As for the preclassification accuracy, the GNR-PCH-FLICM could correctly determine the 98.96% of pixels. In general,

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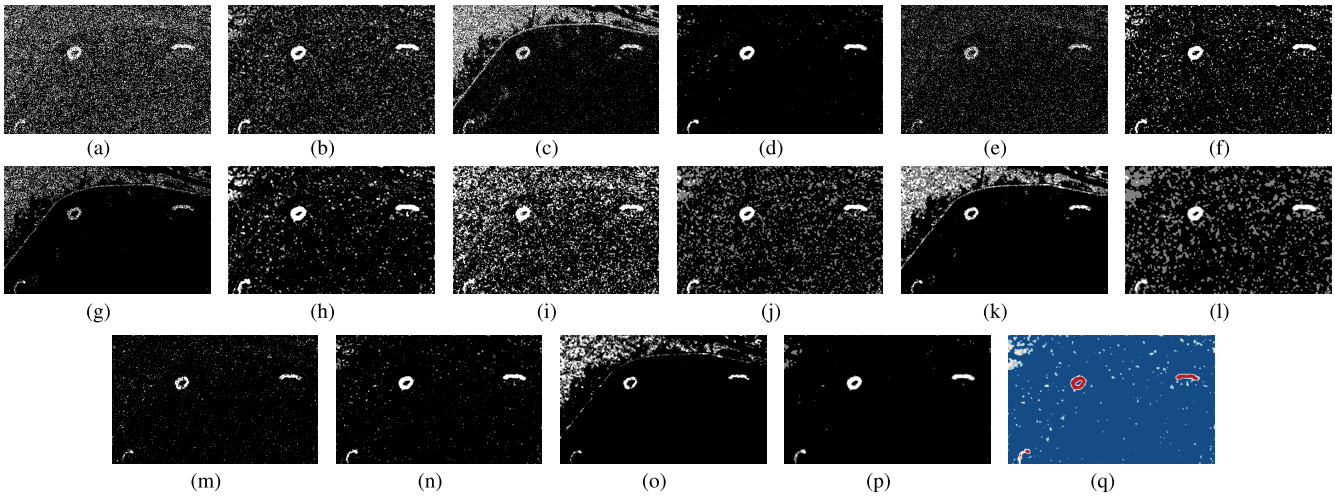

Fig. 17. Visualization of various preclassification results on Coast dataset (a) LR-FCM3, (b) LMR-FCM3, (c) NR-FCM3, (d) GNR-FCM3, (e) LR-H-FCM, (f) LMR-HFCM, (g) NR-HFCM, (h) GNR-HFCM, (i) LR-FLICM3, (j) LMR-FLICM3, (k) NR-FLICM3, (l) GNR-FLICM3, (m) LR-MHFLICM, (n) LMR-MHFLICM, (o) NR-MHFLICM, (p) PCH-FLICM (after merging HUPs into UPs), and (q) PCH-FLICM.

the proposed preclassification method generated the highest number of training samples and the least amount of pixels to be predicted, so that the lower limit of the change detection accuracy of this method was higher than that of other methods. Besides, the higher ICV and SA were more conducive to training more accurate and effective models. Most pixels were correctly labeled, and only a small proportion of pixels were fed into the deep models. The better SA also push the network to produce more accurate predictions, resulting in a higher upper limit of the final change detection accuracy.

D. Impact of Input Patch Size

The size of input patches directly affects the amount of difference information fed into the network, and therefore greatly influences the final change detection result. On the one hand, patches that are too small cannot fully reflect the differences between bitemporal image patches and are susceptible to the interference of isolated noise. On the other hand, patches that are too large may contain heterogeneous pixels, which may hinder the accurate detection of changed areas. To find the optimal size of input patch, a comparative experiment was conducted on the five dataset with varying patch sizes ranging from 3×3 to 13×13. Detailed results are shown in Fig. 18 and Table IX. The optimal patch size for Daxing-I, Daxing-II, Yellow River, and Ottawa datasets was 5×5, while for the Coast dataset was 7×7. It can be seen that for the proposed CBNTNet, the selected patch size should be neither too large nor too small; typically, a moderate patch sizes of 5×5 or 7×7 may be more appropriate.

E. Analysis of Varying Weight Parameters

The proposed CBNTL combines the respective strength of CBAFL to handle imbalance samples with the noise resistance of MAEL. The weight parameter β was used to adjust the trade-off between MAEL and CBAFL to determine the optimal setting of CBNTL. As shown in Fig. 19, we used KC as the evaluation metric and set the β from 0.1 to 0.9 with

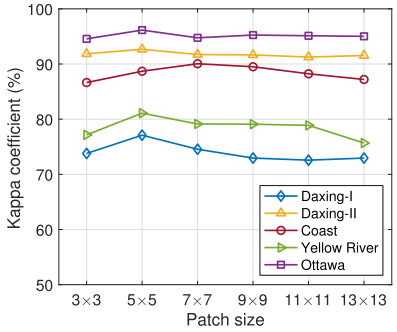

Fig. 18. Comparisons of KCs with varying patch sizes.

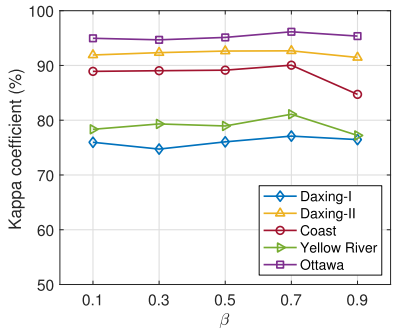

Fig. 19. Comparisons of KCs with β.

an interval of 0.2 for performance comparisons on the five datasets. It worth mentioning that the closer β is to 0, the closer the loss function is to that of MAEL, and the closer β is to 1, the closer the loss function is to that of CBAFL. Although there was some randomness in the training and prediction of the network, the β curves on all the datasets showed similar shapes. From Table X, with the increase of β, the OE value gradually decreased. When it tended to 0.7, the OE reached the lowest value. With further increase of β, the number of OE began to grow, and the KC value started to drop. We can observe that the incorporation of CBAFL

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TABLE IX
CHANGE DETECTION RESULTS WITH VARYING PATCH SIZES

Dataset	Patch	FP	FN	OE	PCC(%)	KC(%)
Daxing-I	3×3	**358**	783	1141	99.29	73.78
	5×5	486	583	**1069**	**99.33**	**77.10**
	7×7	596	612	1208	99.25	75.55
	9×9	841	**529**	1370	99.14	72.96
	11×11	797	570	1367	99.15	72.57
	13×13	730	594	1324	99.17	72.95
Daxing-II	3×3	**218**	683	901	98.31	91.85
	5×5	431	407	**838**	**98.43**	**92.67**
	7×7	350	580	930	98.26	91.72
	9×9	299	632	931	98.25	91.65
	11×11	631	**382**	1013	98.10	91.26
	13×13	434	525	959	98.20	91.54
Coast	3×3	**38**	286	324	99.74	86.64
	5×5	102	190	292	99.77	88.69
	7×7	91	**167**	**258**	**99.80**	**90.05**
	9×9	75	193	268	99.79	89.50
	11×11	60	234	294	99.77	88.23
	13×13	61	256	317	99.75	87.20
Yellow River	3×3	714	1086	1800	98.61	77.16
	5×5	**680**	847	**1527**	**98.82**	**81.09**
	7×7	759	926	1685	98.70	79.13
	9×9	791	907	1698	98.69	79.10
	11×11	823	900	1723	98.67	78.88
	13×13	1391	**755**	2146	98.34	75.68
Ottawa	3×3	324	1117	1441	98.58	94.56
	5×5	607	**438**	**1045**	**98.97**	**96.15**
	7×7	582	829	1411	98.61	94.75
	9×9	**291**	971	1262	98.76	95.25
	11×11	386	916	1302	98.72	95.12
	13×13	410	919	1329	98.69	95.02

TABLE X
CHANGE DETECTION RESULTS WITH VARYING β

Dataset	β	FP	FN	OE	PCC(%)	KC(%)
Daxing-I	0.1	510	611	1121	99.30	75.97
	0.3	288	792	1080	99.33	74.73
	0.5	412	669	1081	99.32	76.05
	0.7	486	**583**	**1069**	**99.33**	**77.10**
	0.9	**286**	733	1019	99.36	76.46
Daxing-II	0.1	**403**	513	916	98.28	91.91
	0.3	414	457	871	98.37	92.34
	0.5	455	**389**	844	98.42	92.63
	0.7	431	407	**838**	**98.43**	**92.67**
	0.9	350	607	957	98.21	91.46
Coast	0.1	68	212	280	99.78	88.92
	0.3	77	202	279	99.78	89.04
	0.5	74	202	276	99.78	89.14
	0.7	91	**167**	**258**	**99.80**	**90.05**
	0.9	**52**	316	368	99.71	84.72
Yellow River	0.1	848	920	1768	98.63	78.34
	0.3	972	769	1741	98.65	79.32
	0.5	1033	**754**	1787	98.62	78.95
	0.7	**680**	847	**1527**	**98.82**	**81.09**
	0.9	920	949	1869	98.55	77.22
Ottawa	0.1	308	1030	1338	98.68	94.96
	0.3	**236**	1169	1405	98.62	94.68
	0.5	497	812	1309	98.71	95.12
	0.7	607	**438**	**1045**	**98.97**	**96.15**
	0.9	488	757	1245	98.77	95.36

was conducive to improving the model's learning ability for imbalanced samples with insignificant effect on noise. On the contrary, the MAEL had some tolerance on noisy labels, but it was harder to converge with imbalanced samples. The CBNTL unified the advantage of both losses, improved the change detection accuracies, and further proves its effectiveness.

TABLE XI
CHANGE DETECTION RESULTS WITH DIFFERENT TYPES OF LOSS FUNCTIONS

Dataset	Loss	FP	FN	OE	PCC(%)	KC(%)
Daxing-I	BCEL	**283**	761	1044	99.35	75.72
	CBAFL	298	717	**1015**	**99.37**	76.70
	MAEL	494	630	1124	99.30	75.73
	CBNTL	486	**583**	1069	99.33	**77.10**
Daxing-II	BCEL	372	563	935	98.25	91.70
	CBAFL	381	516	897	98.32	92.06
	MAEL	**262**	985	1247	97.66	88.51
	CBNTL	431	**407**	**838**	**98.43**	**92.67**
Coast	BCEL	129	**165**	294	99.77	88.83
	CBAFL	**61**	215	276	99.78	89.03
	MAEL	63	251	314	99.75	87.36
	CBNTL	91	167	**258**	**99.80**	**90.05**
Yellow River	BCEL	864	975	1839	98.58	77.37
	CBAFL	889	897	1786	98.62	78.28
	MAEL	918	858	1776	99.70	78.57
	CBNTL	**680**	**847**	**1527**	**98.82**	**81.09**
Ottawa	BCEL	563	884	1447	98.57	94.60
	CBAFL	**268**	1039	1307	98.71	95.07
	MAEL	399	1238	1637	98.39	93.81
	CBNTL	607	**438**	**1045**	**98.97**	**96.15**

F. Analysis of Different Types of Loss Functions

To further verify the effectiveness of the proposed CBNTL, we compared its performance with the standalone BCEL, CBAFL, and MAEL. As presented in Table XI, the proposed CBNTL achieved the highest accuracies and KCs on all five datasets. Using only MAEL resulted in the lowest detection accuracy on the Daxing-II, Coast, and Ottawa datasets, with a significant decrease. This indicates that the imbalance of samples indeed affected the model's prediction. Besides, using only focal loss produced the second best results on all datasets except the Yellow River dataset. It can be seen that the CBNTL based on CBAFL and MAEL possessed stronger noise suppression ability, helped the network to avoid overfitting to imbalanced samples, and demonstrated its superiority to using a single loss function.

V. CONCLUSION

In this article, we have developed an automated SAR change detection framework based on enhanced edge information and prototype constrained clustering, referred to as EIPCNet. It is capable of effectively suppressing noise interference and identifying changes between bitemporal image pair. Advancement of the proposed method are primarily in three aspects. First, we have proposed the GNR difference extraction algorithm to enhance the differential expression capability of the difference map for edge information while suppressing inherent speckle noise. Second, to address the sample imbalance problem, the PCH-FLICM clustering algorithm have been developed for preclassification, facilitating the selection of more balanced and accurate changed and unchanged samples. Finally, the CBNTNet has been constructed to further balance model learning, enhance noise resistance, and achieve reliable change detection results. Comprehensive experimental results and quantitative analysis markedly illustrate the efficiency and effectiveness of the proposed EIPCNet in change detection. Future work will entail exploring semantic change detection with improved interpretability.

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